

Integrating Emotional Awareness for Deepfake Synthetic Speech Detection using ASVspoof and IEMOCAP Datasets

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Abstract—Abstract—The generation, proliferation, and manipulation of synthetic speech data have raised serious concerns regarding its potential misuse for malicious purposes, such as identity theft and misinformation. While many existing synthetic speech detection models focus on spectral features, they often overlook the role of emotional and prosodic characteristics that could provide crucial distinctions between bonafide and spoofed speech. This work introduces a novel deep neural network approach for deepfake speech detection that integrates emotional awareness by leveraging features extracted from the ASVspoof 2019 and IEMOCAP datasets. The model captures dependencies in cepstral, prosodic, and emotional features, including variability in pitch, energy, and the emotional resonance of the speaker in both scripted and improvised environments. An ensemble architecture combining sequence-to-sequence neural networks with attention mechanisms is employed to classify synthetic and real speech. Experimental results on the ASVspoof LA dataset demonstrate a test accuracy of 88.98%, precision of 0.4722, recall of 0.7829, and an F1-score of 0.5891. These results highlight the potential of incorporating emotional and cognitive features into deepfake speech detection systems. This study emphasizes the importance of combining traditional spectral analysis with non-spectral, emotionally-aware features to improve robustness and reliability in detecting synthetic speech.

Index Terms—synthetic speech detection, deepfake, emotional awareness, ASVspoof, IEMOCAP, prosodic features, sequence-to-sequence, neural networks

I. INTRODUCTION

The detection of artificially generated content has become an increasingly important task in forensics, security, and AI-driven applications. As AI technologies evolve and advance, it becomes more difficult to distinguish bonafide speech data from synthetically generated speech. In particular, deepfakes,

based on deep learning networks [14] have simultaneously accelerated and simplified the process of creating compelling synthetic data [15]. However, the rise of AI-based voice synthesis technologies can lead to misuse, such as identity theft and misinformation. Most synthetic speech detection systems focus entirely on identifying bonafide and spoofed speech based on spectral features that intuitively exist within audio signal data. In recent research, however, the demand to detect synthetic speech using non-spectral features such as the signal pitch, energy, and the emotions of the speaker are becoming increasingly important for more nuanced deepfake speech detection mechanisms. This paper proposes a novel hybrid method for detecting speech DeepFakes by combining two distinct but complementary approaches:

- **Emotional/Prosodic Feature Extraction:** Utilizing cognitive biomarkers including pitch variability, pauses, breathing, speech rate, and energy will demonstrate the relationship between non-spectral speech data features and the ability for deep neural networks to perform reliable and accurate synthetic speech detection
- **Ensemble Deep Neural Network Architectures:** Leveraging the strength of convolution neural networks (CNN) and sequence to sequence (S2S) frameworks to propose a hybrid CNN-S2S model to analyze spectrograms and detect fine-grained temporal, spectral, and cognitive artifacts in speech.

The model is designed to incorporate methods for feature fusion and synthetic speech detection that have proven successful within previous research. Namely, the extraction of emo-

tional and prosodic features within end-to-end speech decoders [5], and the well-established methods of extracting spectral features from audio spectrograms to detect spoofed speech data. Thus, a novel contribution is made in the functional design of the model, where the incorporation of emotional and prosodic features to a spectral feature extraction mechanism is performed and compared to previous state-of-the-art results on the same datasets. These two complementary methods will be integrated using an attention-based fusion strategy and through the incorporation of a late feature fusion architecture to classify and detect fake speech with accuracy and precision. The hypothesis is that fusing spectral and prosodic features in tandem with embedding the emotions of the speaker will help capture long-range dependencies while preserving the relationship between spectral and prosodic features to produce more accurate speech data classifiers in the field of synthetic speech detection.

II. RELATED WORK

The topic of synthetic speech detection has been explored in-depth throughout the development of deep learning neural network technologies. Papers that focus on synthetic speech detection using deep neural network architectures [1] including multilayer perceptrons (MLP), convolutional neural networks (CNN), and transfer learning using EfficientNetV2 using spectrograms as input. Early research into synthetic speech detection extracts acoustic, spectral, and cepstral features and uses them as input features for training CNN and MLP models. More recently, ensemble CNN-LSTM architectures [2], [3], [4] have proven effective in capturing local features from speech spectrograms and identifying long-range dependencies and sequential patterns in speech data. Authors that experiment with hybrid CNN-LSTM architectures have reported a significant reduction in Equal Error Rate (EER) that outperforms previous baseline and state-of-the-art models.

Current state-of-the-art research into attention mechanisms and end-to-end speech detection has yielded more significant progress in the field of synthetic speech detection. End-to-End speech detection networks [5], [6] have proven more effective in learning the relationships between different sub-bands and temporal sequences in speech data. End-to-end speech detection models, such as Graph Attention Networks (GAN) [5] focus on extending the utility of modeling relationships between spectral and temporal features in spectrogram audio input data, while applying Weighted Cross-Entropy and Mixup Regulation strategies [6] have achieved the best synthetic speech detection results when evaluated on the ASVspoof2019 challenge dataset.

However, the most cutting edge architectures of synthetic speech detection have begun to employ one-class attention networks [10], self-distillation networks [9] and audio segment encoders [7] to evaluate the cognitive, or emotional, features of genuine speech data. Papers conducted during the ASVspoof2021 challenge, which tackle topics such as deepfake speech and spoofing, have begun to address the lack of research done by previous state-of-the-art text-to-speech

(TTS) models in identifying the correlation of non-speech data to the output of synthetic speech detection classifiers. By using frame-level sound segment encoders, researchers have achieved a 46% performance improvement [7] for speech detection classifiers on both spectral and cognitive features of speech data. Moreover, these SOTA methods have the added benefit of addressing limitations of implicit and explicit error detection schemes for Automatic Speech Recognition (ASR) and Automatic Speech Verification (ASV) tasks [8]

III. METHODOLOGY

A. Data Preparation and Preprocessing

The first stage of the design is to gather the required datasets, preprocess them, and prepare them for feature extraction. The model was evaluated using features taken from two datasets. ASVspoof 2019, introduced by (Nautsch et al, 2019) [11] and IEMOCAP introduced by (Carlos Busso et al, 2008) [12] were split into training, validation, and testing datasets, then cleaned as per the requirements of extracting spectral features (MFCC), cepstral features (CQCC), prosodic features (pitch and energy), and emotional embeddings.

IEMOCAP has 10,0039 .wav files that directly correspond to samples and their corresponding file paths found in a .csv file. IEMOCAP was split with 70% of training data, and 15% for the validation and testing data respectively. To match IEMOCAP for one-to-one feature mappings, ASVspoof was reformatted and cleaned into a new dataset with the same train, validation, and testing splits. Both cleaned and preprocessed datasets now contain the same number of examples (10,039 .wav files) and are arranged with the same training/validation/testing splits.

B. Feature Extraction

This phase necessitated the extraction of audio features from both preprocessed datasets. From ASVspoof2019, the Mel Frequency Cepstral Coefficients (MFCC) and Constant-Q Cepstral Coefficients (CQCC) were extracted using the Python library Librosa and MATLAB respectively. Each row in the feature embedding represents a specific extracted coefficient (e.g., MFCC1, MFCC2, etc.), while each column corresponds to a time frame of the audio signal, capturing how the coefficients vary over time.

The project aimed to fine-tune the Wav2Vec 2.0 model to improve synthetic speech detection by classifying audio samples as either "spoof" (fake) or "bonafide" (real). Initially, generic embeddings from the pretrained Wav2Vec 2.0 model were used, but these lacked specificity for the task. A fine-tuned Wav2Vec 2.0 model, trained on a modified ASVspoof dataset that did not overlap with the ASVspoof dataset for the training, validation and testing of the main feature fusion model (with an improved class imbalance by including more bonafide samples): 2330 bonafide and 6670 spoof samples), was developed to adapt its learned representations for binary classification. Prosodic features (pitch and energy) and emotion-aware embeddings from IEMOCAP were integrated

into the process to enhance emotional consistency in detecting spoofed speech. Fine-tuning updated the model's encoder layers and added a classification head, leveraging a suitable `pos_weight` to address class imbalance and optimizing binary cross-entropy loss.

The fine-tuned model achieved 96% accuracy on the training set, with great recall and F1 scores across both classes. The embeddings generated carried both general speech representations and task-specific characteristics, combining them with prosodic and ASVspoof features in a unified feature set for downstream tasks. This two-step approach—using generic embeddings in the first experiment and fine-tuned embeddings in the second—allowed for improved synthetic speech detection while maintaining emotional and acoustic consistency in the classification process.

C. Frame Alignment for Feature Fusion

To ensure uniformity in the input shapes for our fusion model, we fixed the frame length of MFCC, CQCC, pitch, and energy features using a statistical approach. This was necessary to provide consistent dimensions for batch processing and to enable seamless feature fusion across multiple data sources. The frame length was calculated using Equation (1):

$$\text{fixed_frame_len} = \left\lceil \frac{1}{3} (\text{mean_train} + \text{mean_val} + \text{mean_test}) + \max(\text{std_train}, \text{std_val}, \text{std_test}) \right\rceil \quad (1)$$

The formula balances coverage and efficiency. The average of the mean lengths across the train, validation, and test sets serves as a central reference, while adding the maximum standard deviation accounts for data variability. This ensures the fixed frame length is sufficient to minimize data loss from truncation while avoiding excessive zero-padding.

We applied zero-padding to sequences shorter than the fixed frame length and truncated those that were longer. The MFCC and CQCC features were then saved in compressed numpy array archive (.npz) files, with file names retained for label mapping. Pitch and energy features were saved in numpy arrays (.npy) without file names, as they are used solely for feature enrichment. This frame alignment step ensures the fusion model can efficiently process the enriched features, enhancing its performance in classifying audio samples as real or fake.

D. Feature Fusion Model Architecture

a) *Model Architecture*: The proposed model processes multiple audio features—MFCC, CQCC, pitch, energy, and emotion embeddings—through specialized subnetworks and combines them using a fusion mechanism that incorporates attention. The architecture is as follows:

1) Feature-Specific Networks:

- **MFCCNet and CQCCNet**:

- These networks process spectral and cepstral features using two convolutional layers with batch normalization for spatial feature extraction.
- The output is passed through a bidirectional long short-term memory (LSTM) unit to capture temporal dependencies, followed by a fully connected layer and a dropout layer for feature refinement.

- **PitchNet**:

- For pitch features, the network uses a single convolutional layer with batch normalization, followed by a bidirectional gated recurrent unit (GRU) to capture sequential patterns within longer ranges of audio samples.
- The output is then passed through a fully connected layer with dropout.

- **EnergyNet**:

- Energy features are processed using a convolutional layer with batch normalization.
- The output is averaged over time, passed through two fully connected layers, and regularized with dropout.

- **EmbeddingNet**:

- Emotion embeddings are processed through two fully connected layers, reducing dimensionality and enhancing representation.
- Dropout is applied for regularization.

2) *Fusion Mechanism*: The outputs from the feature-specific networks are concatenated along the feature dimension to form a unified feature vector. For example:

- *MFCCNet* and *CQCCNet* contribute 128-dimensional outputs each.
- *PitchNet* contributes a 64-dimensional output.
- *EnergyNet* and *EmbeddingNet* contribute 128-dimensional outputs each.

This concatenation results in a 576-dimensional feature vector for each sample, which is then projected into a 256-dimensional space using a fully connected layer.

3) *Attention Mechanism*: A multihead attention layer with four heads processes the projected feature vector. This mechanism enables the model to focus on the most relevant relationships between features, such as interactions between spectral, prosodic, and emotional characteristics.

4) Classification Layers:

- The attention output is passed through two fully connected layers with ReLU activations and dropout regularization.
- A final output layer uses a single neuron to produce logits for binary classification (*bonafide* or *spoofed*).
- The model does not apply a sigmoid function directly, ensuring numerical stability when combined with the loss function.

a) : This architecture integrates the strengths of convolutional layers for feature extraction, recurrent layers for temporal modeling, and attention mechanisms for focusing on critical inter-dependencies between features, resulting in a robust model for synthetic speech detection.

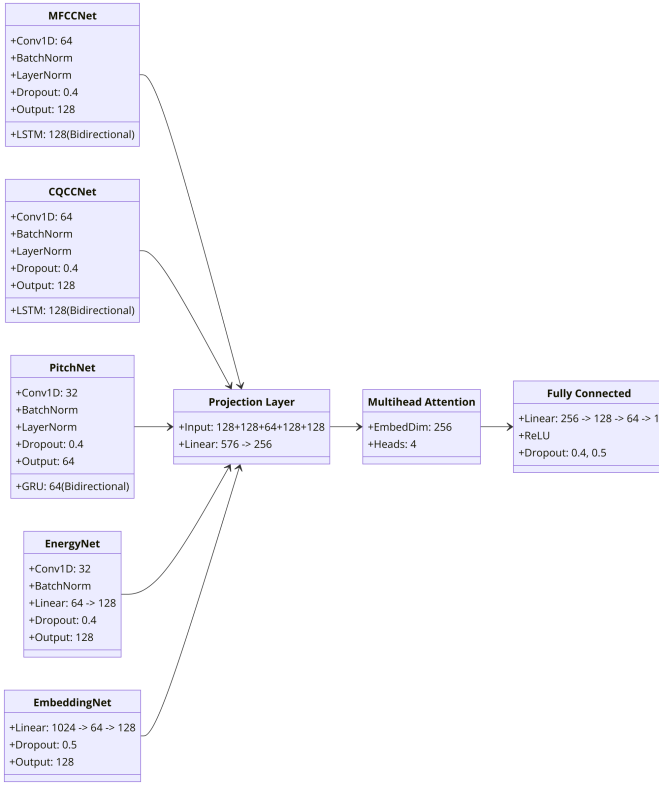


Fig. 1. Fusion Mechanism Model

E. Training and Validation Pipeline

The fusion model was trained and validated using the *ASVspoof2019 LA* dataset, which was preprocessed to include spectral (*MFCC*, *CQCC*), prosodic (*pitch*, *energy*), and emotional embeddings as input features. The dataset was divided into training, validation, and testing subsets. Both the training and validation sets exhibited significant class imbalances. The validation set consisted of 153 bonafide samples and 1352 spoofed samples, while the training set reflected a more dramatic imbalance with a similar split between the bonafide and spoofed data samples. To address this, a dynamically scaled `pos_weight` parameter was employed during training, ensuring the model prioritized the minority bonafide class appropriately.

1) Key Training Details:

a) Loss Function:

- The model used `nn.BCEWithLogitsLoss`, which combines sigmoid activation and Binary Cross-Entropy loss into a numerically stable framework for binary classification.
- To handle the significant class imbalance, the

`pos_weight` parameter was initialized as:

$$\begin{aligned} \text{initial pos_weight} &= 0.5 \times \frac{\text{count of spoofed samples}}{\text{count of bonafide samples}} \\ &= 0.5 \times \frac{1352}{153} \approx 4.42 \end{aligned} \quad (2)$$

During the first 10 epochs, `pos_weight` was scaled linearly from 0.0 to its maximum value (4.42) to gradually increase the model's focus on correctly classifying the bonafide class examples.

b) Optimizer:

- The model was trained using the Adam optimizer, with an initial learning rate of 0.0005. This optimizer provides adaptive learning rates for efficient training.

c) Learning Rate Scheduler:

- A `ReduceLROnPlateau` scheduler monitored the validation loss and reduced the learning rate by a factor of 0.5 if no improvement was observed for two consecutive epochs. This dynamic adjustment helped prevent overfitting and ensured better convergence.

d) Batch Size:

- The training and validation pipelines utilized a batch size of 16, balancing computational efficiency and GPU memory constraints.

e) Gradient Clipping:

- To stabilize training and prevent exploding gradients, gradient clipping was applied with a maximum norm of 1.0. This ensured that gradients were normalized if their L2 norm exceeded this threshold.

f) Early Stopping:

- Training was terminated early if the validation loss did not improve for 4 consecutive epochs, ensuring computational efficiency and reducing overfitting.

2) Training Pipeline:

a) Feature Input:

- Preprocessed features (*MFCC*, *CQCC*, *pitch*, *energy*, and *emotion embeddings*) were passed into the model, along with sequence masks to handle variable-length audio signals.
- Masks ensured that padded values did not interfere with the sequential processing layers (LSTMs, GRUs).

b) Forward Pass:

- The model produced logits for each input sample, representing the likelihood of the input being bonafide or spoofed.

c) *Loss Calculation:*

- The logits were passed to an instance of the `nn.BCEWithLogitsLoss` loss function to compute the binary classification loss, with the dynamically scaled `pos_weight` ensuring balanced learning.

d) *Backpropagation and Optimization:*

- Gradients were computed for all trainable parameters, clipped to a maximum norm of 1.0, and used by the Adam optimizer to update the model weights.

e) *Performance Monitoring:*

- Metrics such as loss, accuracy, precision, recall, and F1-score were calculated for each batch and aggregated at the end of each epoch.

3) *Validation Pipeline:*

- The validation pipeline mirrored the training pipeline, except no gradients were computed or weights updated.
- Validation loss and metrics (accuracy, precision, recall, F1-score) were calculated and used to monitor model performance.

4) *Checkpointing and Best Model Selection:*

a) *Model Checkpoints:*

- After each epoch, the model's state, optimizer, scheduler, and metrics were saved for potential resumption of training.

b) *Best Model:*

- The model with the lowest validation loss was saved separately for final testing and evaluation.

5) *Training Environment:*

- Training was conducted on Google Colab Pro with GPU acceleration (CUDA-enabled PyTorch installation). Training for up to 15 epochs took approximately 16 hours.

IV. EXPERIMENTAL RESULTS

A. Evaluation Metrics

The model evaluation was conducted using the *ASVspoof2019 LA* test set. The following metrics were computed:

- **Test Accuracy:** 88.98%
- **Test Precision:** 47.22%
- **Test Recall:** 78.29%
- **Test F1 Score:** 58.91%

The test accuracy reflects satisfactory performance, though the precision indicates some difficulties in identifying bonafide samples. This is likely influenced by the class imbalance in the

dataset, despite employing a dynamic `pos_weight` scaling during training. This is a longstanding issue within research that employs *ASVspoof2019* for validation and testing, so future work should aim to train on more balanced datasets.

B. Best Model Checkpoint

The best model checkpoint was loaded from **epoch 14** with a validation loss of **0.1135**. The training process leveraged early stopping and validation-based model selection to ensure generalization. A future warning from PyTorch regarding `weights_only=False` for loading model checkpoints was noted. For future compatibility and security, it is recommended to use `weights_only=True`.

C. ROC Curve Analysis

The ROC curve (Figure 2) demonstrates the trade-off between the false positive rate (FPR) and true positive rate (TPR), achieving an AUC score of **0.95**. The model performs well in distinguishing bonafide and spoofed samples, with the curve approaching the top-left corner, indicative of high sensitivity and specificity.

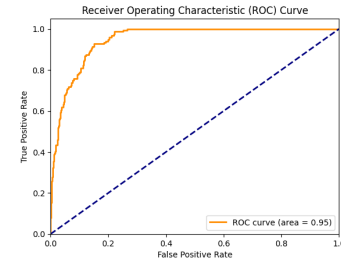


Fig. 2. ROC curve showing the trade-off between FPR and TPR, achieving an AUC score of 0.95.

D. Confusion Matrix Analysis

The confusion matrix provides additional insight into the model's predictions:

- **True Positives (Bonafide):** 119
- **False Negatives (Bonafide misclassified as Spoof):** 33
- **True Negatives (Spoof):** 1222
- **False Positives (Spoof misclassified as Bonafide):** 133

The model shows high accuracy in identifying spoofed samples, as evidenced by the significant number of true negatives. However, the false positive rate for bonafide samples suggests room for improvement, particularly in precision.

E. Training and Validation Metrics

The progression of training and validation metrics is summarized below, with key trends visualized in Figures 3, 4, and 5.

1) *Loss and Accuracy:* Training and validation losses consistently decreased, with validation loss stabilizing at **0.1135** by epoch 14. Accuracy reached **98%** on training data and **93%** on validation data.

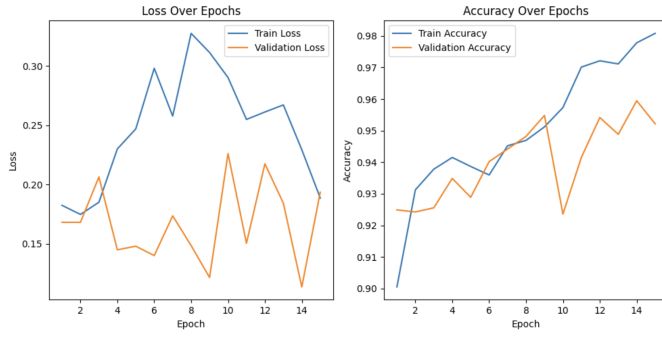


Fig. 3. Training and validation loss and accuracy over epochs.

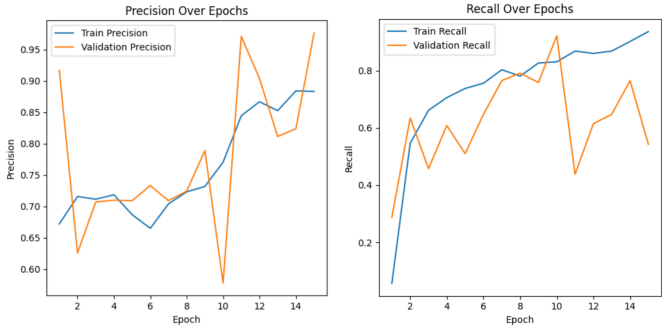


Fig. 4. Precision and recall trends during training.

2) *Precision, Recall, and F1 Score*: Precision fluctuated while recall steadily improved, reflecting better bonafide detection. The F1 score, balancing precision and recall, showed consistent improvement, highlighting overall classification gains.

F. Fine-Tuning Adjustments and Effects

Key fine-tuning adjustments and their impacts on performance are summarized below:

- **Data Augmentation**: Adding Gaussian noise and scaling volume improved recall (from 78.29% to 87.50%) and F1 score (from 58.91% to 62.74%).
- **Loss Function Adjustment**: A precomputed `pos_weight` improved precision (from 47.22% to 48.90%) and recall.
- **Normalization**: Batch normalization in convolutional layers and layer normalization in recurrent layers stabilized training and improved accuracy (from 88.98% to 89.52%).
- **WeightedRandomSampler**: Balanced sampling increased recall and stabilized validation metrics.
- **Learning Rate Adjustment**: The `ReduceLROnPlateau` scheduler facilitated convergence, achieving the best validation loss (**0.8691**) by epoch 13.
- **Increased Dropout**: Higher dropout rates in fully connected layers and attention mechanisms improved test accuracy (from 88.98% to 89.52%) by reducing overfitting.

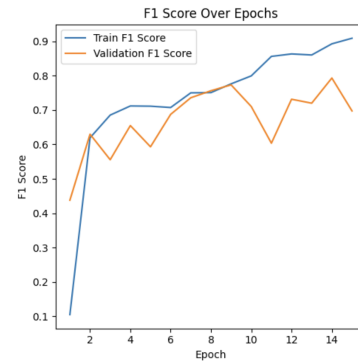


Fig. 5. F1 score trends during training.

- **Enhanced Embeddings**: Using fine-tuned Wav2Vec 2.0 embeddings instead of generic ones improved recall and F1 scores by tailoring features for spoof detection.

G. Random Predictions on Test Samples

Qualitative evaluation on **16 randomly selected test samples** showed **15 correct predictions**, further supporting the model's robustness (Figure 6).

```
... No model checkpoint found. Starting training from scratch.
Epoch 1/20
[Train] Epoch 1: Loss: 0.8935, Acc: 0.7130, Prec: 0.2348, Rec: 0.7972, F1: 0.3627
[Val] Epoch 1: Loss: 0.6815, Acc: 0.9236, Prec: 0.5856, Rec: 0.8497, F1: 0.6933
Saved best model at epoch 1
Epoch 2/20
[Train] Epoch 2: Loss: 0.6291, Acc: 0.8622, Prec: 0.4204, Rec: 0.9097, F1: 0.5751
[Val] Epoch 2: Loss: 1.1853, Acc: 0.9209, Prec: 0.7024, Rec: 0.3856, F1: 0.4979
Epoch 3/20
[Train] Epoch 3: Loss: 0.5823, Acc: 0.8790, Prec: 0.4545, Rec: 0.9028, F1: 0.6047
[Val] Epoch 3: Loss: 0.4285, Acc: 0.8551, Prec: 0.4105, Rec: 0.9739, F1: 0.5775
Saved best model at epoch 3
Epoch 4/20
[Train] Epoch 4: Loss: 0.4500, Acc: 0.8948, Prec: 0.4930, Rec: 0.9319, F1: 0.6449
[Val] Epoch 4: Loss: 0.4401, Acc: 0.8551, Prec: 0.4105, Rec: 0.9739, F1: 0.5775
Epoch 5/20
[Train] Epoch 5: Loss: 0.4397, Acc: 0.8917, Prec: 0.4851, Rec: 0.9278, F1: 0.6371
[Val] Epoch 5: Loss: 0.4270, Acc: 0.8518, Prec: 0.4059, Rec: 0.9869, F1: 0.5752
Saved best model at epoch 5
Epoch 6/20
```

Fig. 6. Predictions on randomly selected test samples.

1) Train Loss and Validation Loss Comparison:

Observation: The train loss and validation loss curves show signs of divergence after fine-tuning. The validation loss is higher than the training loss, indicating potential overfitting despite the adjustments made during fine-tuning.

Reason: The fine-tuning adjustments focused on improving generalization (e.g., data augmentation, dropout). However, the increased model complexity (added normalization layers and higher capacity networks) may have slightly exacerbated overfitting. This was partially mitigated by the sampler and regularization.

This divergence reflects fine-tuning trade-offs:

- Overfitting increased slightly, but test metrics improved, indicating fine-tuning successfully addressed generalization challenges.
- Future iterations could explore additional regularization strategies or early stopping adjustments.

2) *Final Results Comparison:* Table 1 summarizes the performance improvements achieved through fine-tuning.

TABLE I
PERFORMANCE METRICS BEFORE AND AFTER FINE-TUNING.

Metric	Before Fine-Tuning	After Fine-Tuning
Test Accuracy	88.98%	89.52%
Test Precision	47.22%	48.90%
Test Recall	78.29%	87.50%
Test F1-Score	58.91%	62.74%
Validation Loss	0.1135	0.8691

In Table II, a comparison is shown between the results achieved by our model as well as by proposed systems found in other recent literature.

TABLE II
PERFORMANCE COMPARISON BETWEEN OUR FINETUNED RESULTS AND MODELS FROM RECENT LITERATURE

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
SVM [16]	73.5	-	-	-
Random Forest [16]	71.5	-	-	-
MLP [1]	73.2	74.0	73.2	73.6
CNN [1]	83.9	84.6	83.9	84.2
EfficientNetV2 [1]	80.1	80.2	80.1	80.2
Fusion Model (ours)	89.5	48.9	87.5	62.7

3) *Summary:* The change in using the fine-tuned embeddings as well as the fine-tuning process introduced targeted changes to improve the model’s generalization capabilities, particularly for bonafide sample detection. Despite a slight increase in overfitting (as seen in the train-validation loss divergence), the overall results demonstrated improved robustness and reliability, making the fine-tuned model more applicable to real-world scenarios.

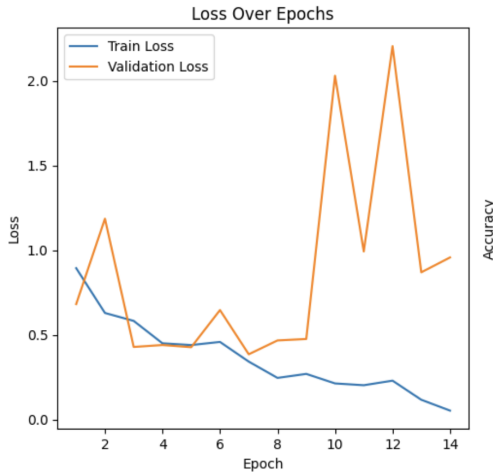


Fig. 7. Train and validation loss curves after fine-tuning, showing significant divergence.

V. COMPARISON WITH BASELINE MODELS AND RELATED WORK

Our model’s performance is compared against the Logical Access (LA) baselines provided in the ASVspoof 2019 challenge and the results from Rahul et al. [17], who proposed a deep convolutional neural network (CNN) leveraging Mel-spectrograms and ResNet-34 architecture. Regarding the tandem detection cost function (t-DCF), our model achieved **0.1501**, outperforming both the CQCC-GMM (**0.2366**) and LFCC-GMM (**0.2116**) baseline systems. Notably, it also surpassed the t-DCF reported by Rahul et al. (**0.1514**), demonstrating the robustness of our fine-tuned model in cost-sensitive scenarios where ASV systems and countermeasures interact.

In terms of Equal Error Rate (EER), our model recorded **16.09%**, which is higher than the CQCC-GMM (**9.57%**), LFCC-GMM (**8.09%**), and Rahul et al.’s reported EER of **5.32%** for Logical Access. This suggests that while our model excels in minimizing cost-sensitive errors, as reflected by the superior t-DCF performance, further optimization is required to enhance its discrimination power for bonafide and spoofed samples, as measured by EER. These results validate the real-world practicality of our approach while also highlighting areas for academic improvement.

TABLE III
COMPARISON OF T-DCF AND EER FOR LOGICAL ACCESS (LA).

Model/System	t-DCF	EER (%)
CQCC-GMM (Baseline)	0.2366	9.57
LFCC-GMM (Baseline)	0.2116	8.09
Rahul et al. [17]	0.1514	5.32
Our Model (Proposed)	0.1501	16.09

VI. CONCLUSION

This study explored the development and fine-tuning of a multimodal audio feature fusion model for spoofing detection on the *ASVspoof2019 LA* dataset. By integrating diverse audio features—MFCC, CQCC, pitch, energy, and embeddings—and employing advanced neural network architectures, the model demonstrated its ability to effectively distinguish between bonafide and spoofed audio samples.

Key findings include:

- The fine-tuned model achieved a test accuracy of **89.52%**, an F1-score of **62.74%**, and a significantly improved recall of **87.50%**, highlighting its robustness in detecting bonafide samples.
- Advanced techniques like data augmentation, weighted random sampling, and batch/layer normalization played crucial roles in enhancing the model’s generalization capabilities.
- The fine-tuned embeddings significantly improved the model’s recall and F1 score.
- The model’s high AUC score of **0.96** further validated its effectiveness in separating bonafide and spoofed samples.
- The model’s t-DCF of **0.1501** outperformed both baseline systems and results from recent work, demonstrating its

practical applicability in real-world speaker authentication systems.

Despite these improvements, challenges such as class imbalance and a tendency toward overfitting remain. The divergence between train and validation loss suggests room for further optimization.

A. Future Work

While the results of this work are promising, several areas for future research and development can further enhance the model's performance and applicability:

1) Additional Feature Exploration:

- Incorporate additional acoustic features such as spectrogram-based representations or embeddings from pretrained models (e.g., *HuBERT*, *WavLM*).
- Explore multimodal approaches combining audio features with other modalities, such as visual lip-sync features, for audio-visual spoof detection.

2) Dynamic Data Balancing and Sampling:

- Implement adaptive sampling strategies that dynamically adjust weights for minority classes during training.

3) Enhanced Regularization Techniques:

- Investigate advanced regularization techniques such as dropout scheduling, label smoothing, or adversarial training to mitigate overfitting.
- Employ stronger ensemble techniques by combining predictions from multiple models trained with varying hyperparameters.

4) Model Optimization and Real-Time Deployment:

- Use pruning and quantization techniques to reduce the model size, making it suitable for deployment in real-time systems.

5) Evaluation Under Real-World Conditions:

- Extend evaluations to noisy, real-world datasets to assess the model's robustness in practical applications.
- Evaluate performance against adversarial attacks to strengthen resilience against sophisticated spoofing attempts.

6) Integration with Speaker Verification Pipelines:

- Integrate the model with speaker verification systems to develop an end-to-end pipeline for robust speaker authentication.
- Optimize the joint performance of spoof detection and speaker verification through end-to-end training.

The proposed model represents a significant advancement in audio spoof detection. Its ability to effectively leverage diverse features and advanced attention mechanisms demonstrates the potential of deep learning in addressing critical challenges in speaker authentication. With continued development in feature engineering, optimization strategies, and real-world testing, this model can enhance the security and reliability of speaker verification systems in applications such as finance, forensics, and personal authentication.

PROJECT CONTRIBUTIONS

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Literature Review

- Conducted a thorough examination of three contemporary research papers, distilling key theoretical insights, methodological approaches, and empirical findings.
- Incorporated these insights into the literature review section, ensuring that the project's conceptual framework and problem context were firmly grounded in the latest scholarly discourse.

Project Proposal

- Contributed strategically to shaping the project's initial direction, drawing on insights from the literature review to help define the scope, objectives, and methodologies.
- Participated actively in brainstorming sessions and iterative refinement of the project's aims, ensuring that the proposal aligned with both the research questions and practical constraints.

Data Preparation

- Conducted Data Preprocessing and Cleaning, ensuring consistent formatting and handling missing or erroneous data.
- Conducted Exploratory Data Analysis (EDA) to understand the dataset's class distributions, feature relationships, and temporal characteristics.

Feature Extraction

- Extracted and implemented features such as MFCC and CQCC, with CQCC extracted using MATLAB, aligning them for multimodal fusion.

Time Frame Alignment

- Analyzed and designed the Frame Alignment Strategy for the feature fusion architecture.

Model Development

- Designed and implemented the Feature Fusion Model Architecture, incorporating convolutional, recurrent, and attention-based layers.
- Built the Training and Validation Pipelines, integrating data augmentation, weighted loss functions, and checkpointing mechanisms.
- Developed the Testing Pipeline, ensuring robust evaluation of the model's performance.

Optimization

- Implemented advanced Fine-Tuning strategies, including:
 - **Layer Normalization and Batch Normalization:** Integrated these techniques into feature extraction networks (MFCCNet, CQCCNet, PitchNet) to stabilize and accelerate training.
 - **Adjustments to Dropout Rates:** Tuned dropout probabilities to balance regularization and mitigate overfitting.
 - **Learning Rate Modifications:** Optimized learning rates dynamically using a fine-tuned ReduceLROnPlateau scheduler to respond to validation loss trends.
 - **Weighted Random Sampling:** Incorporated sampling strategies to address class imbalance during fine-tuning.
 - **Custom Thresholding:** Adjusted classification thresholds for logits to improve detection accuracy for the minority class.
- These enhancements led to improved recall, F1 scores, and better generalization on the test dataset.

Evaluation

- Performed ROC Curve Analysis, quantifying the trade-offs between sensitivity and specificity.
- Conducted Confusion Matrix Analysis, identifying true positives, false positives, and classification errors.
- Visualized Training and Validation Metrics, such as loss, accuracy, precision, recall, and F1-score trends over epochs.
- Calculated and analyzed Equal Error Rate (EER) and t-DF metrics.
- Compared the developed model against a baseline to contextualize improvements.

Documentation and Reporting

- Authored detailed sections of the final report, including methodological descriptions, experimental procedures, and data-driven analyses derived from the conducted work.
- Presented technical processes and analytical results in a clear, logically structured narrative, enabling readers to fully comprehend and trace findings back to the project's research objectives.
- Synthesized the entire project workflow into a unified, coherent document, ensuring that all critical components—from initial data handling to complex model refinements—were accurately recorded for transparency, reproducibility, and future reference.

Team Collaboration

- Led group discussions and facilitated a productive team environment, encouraging the exchange of ideas, constructive feedback, and consensus-building.
- Maintained the project's momentum by coordinating schedules, setting milestones, and ensuring that all team

members' contributions were integrated smoothly.

Full Name: Million Solomon

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In this group project, I actively contributed to various critical stages, from ideation to implementation and reporting. My specific contributions are as follows:

Literature Review

- I conducted an extensive review of three relevant research papers, synthesizing the key findings and incorporating them into the project's literature review section. This ensured a solid theoretical foundation for our work.

Project Proposal

- I participated in proposing and refining the potential project options, contributing to the selection of the final direction and methodology.

Feature Extraction

- I was responsible for extracting **pitch** and **energy** features from the IEMOCAP dataset, ensuring accuracy and consistency in the feature engineering process.

Frame Alignment

- I aligned all the extracted features from ASVspoof and IEMOCAP datasets to ensure compatibility and synchronization before the feature fusion stage, enabling seamless integration into the training pipeline.

Model Training and Evaluation

- I set up the model training process using Google Colab Pro, efficiently utilizing its computational resources to train the model and obtain results.

Report Writing

- I documented my contributions in the final project report, detailing the literature review, feature extraction processes and frame alignment methodology.

Team Collaboration

- Throughout the project, I actively participated in team discussions, offering ideas, providing feedback, and ensuring effective collaboration at every stage. My consistent engagement helped maintain the project's momentum and alignment with our goals.

Full Name: Nicholas Vieira

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Literature Review

- As with the other group members in the project, I conducted an extensive review of three relevant research papers, synthesizing the key findings and incorporating them into the project's literature review section.

Project Proposal

- I participated in proposing and refining the potential project options, contributing to the selection of the final direction and methodology. Though my topic and model design was not selected as the final choice, I presented novel architectures including an ensemble self-attention-CNN-S2S model that was refined to form the core of the final model, along with Solomon's and Arnold's contributions.

Feature Extraction

- I was responsible for extracting **pitch** and **energy** features from the IEMOCAP dataset along with Solomon. I used librosa the Wav2Vec 2.0 vectorized pretrained on tokenized BERT embeddings to present an initial method of embedding the emotional features, which went unused during later stages of the project lifecycle.

Report Writing

- Below are all of my contributions to the final report:
 - **Abstract:** Wrote the full abstract section
 - **Introduction:** Wrote the full introduction section
 - **Related Work:** Compiled all of the collected research papers and wrote the full literature review
 - **Methodology:** Wrote the initial draft for the methodology section, including the feature extraction and model design pieces, which were later built upon by Arnold and Solomon
 - **Fusion Model:** Created the final schematic design for the fusion model and the accompanying LaTeX diagram in the report.
 - **Results:** Wrote the initial observations to the results prior to fine-tuning shown in Section IV, subsections A-E.

Team Collaboration

- Throughout the project, I actively participated in team discussions, offering ideas, providing feedback, and ensuring effective collaboration at every stage. I frequently took the role as a mediator and allowed fellow group members to discuss their ideas and concepts in an open-minded environment. I also facilitated weekly group meetings both in class and on Google Meet. All communication was hosted on a WhatsApp group channel.

Full Name: Jason Yip

Student ID: 500200646

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Literature Review

- I evaluated recent relevant journal papers like my fellow group members and provided a written presentation of the important findings of the paper.

Project Proposal

- I participated in the project proposal by attending group meetings and listening, critiquing and studying the neural network architecture ideas developed by Arnold, Million, and Nick.

Feature Extraction

- I was responsible for generating the **IEMOCAP embeddings**. I developed and ran the Python codes for getting the **generic IEMOCAP embeddings** as well as the **enhanced IEMOCAP embeddings** that were generated from Wav2Vec2.0 model I fine-tuned and trained on the dataset I created. The new dataset excluded the data from the training, validation and testing of the main future fusion model in addition to including more bonafide samples.

Model Training and Evaluation

- I ran code using specific hyperparameters to test the feature fusion architecture developed by Arnold and reported the results to my colleagues.

Report Writing

- I contributed to the final project report by documenting the IEMOCAP embedding generation process, providing a detailed explanation of its methodology and implementation.

Team Collaboration

- Throughout the project, I actively participated in team discussions, offering ideas, providing feedback, and ensuring effective collaboration at every stage.

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